Three-Layer Projection Model: Nested Subjective Spaces R–R_1–R_0 and Middleware Geometry via Central Projection

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Abstract

This paper organizes and integrates the geometric properties of subjective spaces, already established in the central projection series, into a **three-layer structure** R–R_1–R_0 as a middleware paper. It proves no new theorems and confines itself to observing the following structure as a combination of known properties of central projection:

1. For any curvature radius R, the subjective space $S^n(R)$ is defined as a central projection centered at the common origin O of the background coordinate space.
2. Two subjective spaces $S^n(R_1)$ and $S^n(R_0)$ at different scales are arranged as a concentric nested structure by virtue of origin sharing and projection-axis collinearity.
3. The R-direction displacement connecting corresponding points on different subjective spaces is orthogonal to the tangent spaces of all subjective spaces and hence cannot be measured by any intrinsic metric.

This structure provides an **implementation-independent stable interface** for **upper and lower implementation layers** in which each particle may possess its own subjective space as a micro black hole. This paper relies on neither the specific implementation of particle internal structure nor the specific implementation of phase-space dynamics, and its scope of claims is limited to the geometric middleware connecting the two.

Keywords: central projection, subjective space, background coordinate preservation, origin sharing, concentric structure, metric non-contribution of R-direction displacement, middleware geometry

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§1 Scope of Claims and Separation Principle

The claims of this paper are limited to the **organization and integration** of properties established in the central projection series foundational papers [P1], [P2], [P8]. No new mathematical claims are made.

The following elements are **outside the scope of this paper's claims** and are treated only as implementation examples when mentioned in the text:

- Specific implementations of particle internal structure (including the 6-dimensional hyper-rectangle model [W8])
- Phase-space dynamics inside micro black holes (including soliton wave equations)
- Correspondence with individual particles in the Standard Model

The claims of this paper are unaffected even if the above implementations are entirely incorrect, as long as the paper relies on the geometric properties of central projection themselves. This

follows, at a higher layer, the policy of "separation of mathematical substance and interpretation" in [W8].

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§2 Basic Conditions and Constraints

The three-layer structure of this paper is defined under the following conditions.

Condition 1 (Origin Sharing)

All subjective spaces $S^n(R)$, $S^n(R_1)$, $S^n(R_0)$ share the **same origin** O of the background coordinate space $\mathbb{R}^{n+1}$ as the projection center of central projection. This is a structural condition that follows directly from the definition of central projection (§3.1) and is not an additional assumption.

Condition 2 (Sharing of Axis Directions)

For any subjective space with $R > 0$, the **directions of the half-lines** (projection axes) emanating from the origin O of the background coordinate space **are shared**. That is, even when the curvature radius R differs, the directional structure of the projection axes is identical. This also follows directly from the definition of central projection in §3.1 ($R > 0$).

Condition 3 (Non-constraint on Multiplicity)

There is **no upper bound on the number** of general subjective spaces $S^n(R)$ and particle-scale subjective spaces $S^n(R_1)$. Subjective spaces sharing the same origin O may exist infinitely, one for each distinct curvature radius. This follows from the fact that the definition of central projection holds for any value of curvature radius $R > 0$.

Condition 4 (On the Uniqueness of R_0)

In this paper, since $S^n(R_0)$ is treated as the subjective space corresponding to the 4-dimensional spacetime we observe, R_0 is **assumed to be unique**. However, this is **not a constraint** on the geometric claims of this paper. Even if multiple (or infinitely many) subjective spaces corresponding to R_0 existed, the claims of §3–§6 would not break down. The uniqueness of R_0 is a conventional setting reflecting our observational standpoint.

The above conditions are illustrated below.

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§3 Prerequisites: Basic Properties of Central Projection (Cited)

This section contains no new claims. It organizes the basic properties used in subsequent discussion by citing them from the foundational papers.

§3.1 Definition of Central Projection

With the origin O of the background coordinate space $\mathbb{R}^{n+1}$ as the projection center, the central projection onto the hypersphere $S^n(R) \subset \mathbb{R}^{n+1}$ with curvature radius $R > 0$ is defined as

$$P_R : \mathbb{R}^{n+1} \setminus \{O\} \to S^n(R), \quad P_R(\mathbf{x}) = R \cdot \frac{\mathbf{x}}{\|\mathbf{x}\|}$$

([P1] §2). The image $S^n(R)$ of P_R is called the **subjective space of curvature radius** R.

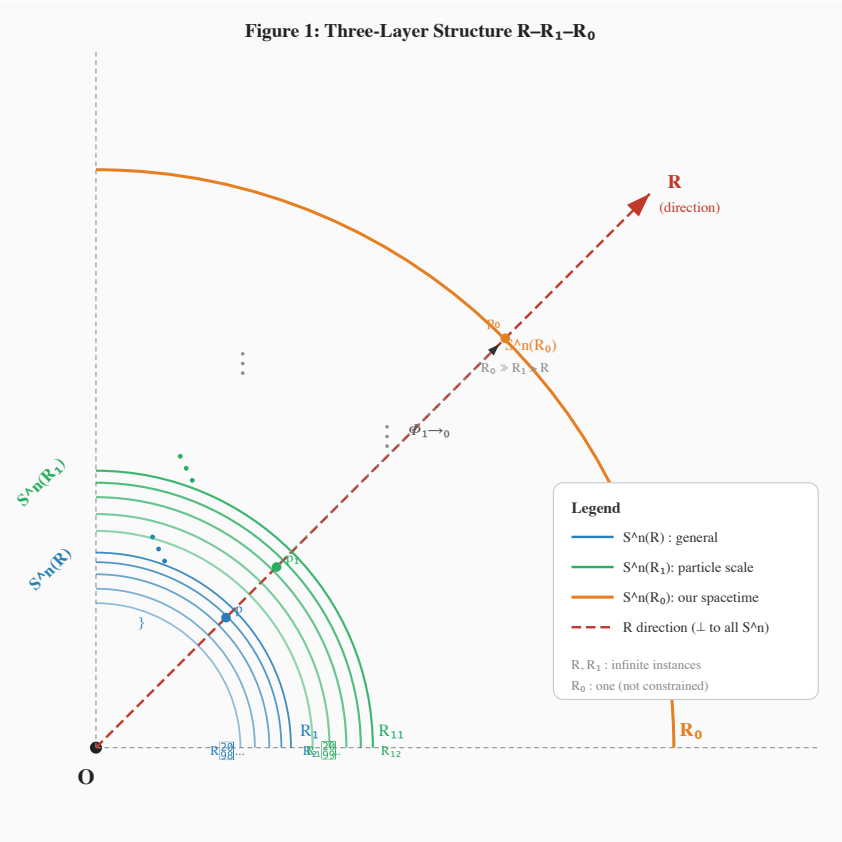

Figure 1: Conceptual diagram of the three-layer structure, drawn as quarter circles sharing the origin O. The blue arc group $S^n(R)$ and the green arc group $S^n(R_1)$ may each have infinitely many instances (Condition 3). The orange arc $S^n(R_0)$ is our subjective space (Condition 4). The red dashed line indicates the R-direction, which is orthogonal to the tangent spaces of all S^n. The points p, p_1, p_0 shown on the 45-degree half-line are corresponding points on the same background coordinate half-line, connected by the map $\Phi_{1\to 0}$ (the 45-degree direction is a drawing convenience; the same correspondence holds for half-lines in any direction).

§3.2 Background Coordinate Preservation

The central projection P_R **preserves background coordinates** ([P1] §3). That is, the original half-line in background coordinates can be uniquely recovered from the projection image via the inverse projection, and this correspondence is regular.

§3.3 Observability within the Subjective Space

The quantities measurable by an observer inside the subjective space $S^n(R)$ are limited to geodesic lengths and winding numbers on $S^n(R)$. In particular, the distance in the radial direction (R-direction) **cannot be perceived** from inside the subjective space ([P2] §4, [P8] §3).

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§4 Definition of the Three-Layer Structure

§4.1 Roles of the Three Symbols

In this paper, the following three symbols are used as curvature radii defined by central projection.

- R: A **generic symbol** denoting the curvature radius of the subjective space $S^n(R)$. The definition parameter of central projection as defined in §3.
- R_1: A **specific instance** denoting the curvature radius of the particle-scale subjective space $S^n(R_1)$. In the upper implementation layer (see §7), it may be interpreted as the curvature radius of a micro black hole.
- R_0: A **specific instance** denoting the curvature radius of our subjective space $S^n(R_0)$. A nearly flat subjective space with $R_0 \to \infty$.

Note: Within the scope of this paper, R is a generic symbol, and R_1 and R_0 are two instances that specialize it. The implementation that identifies $R = R_1$ is presented in §7 (Implementation Examples), but the claims of this paper do not depend on this identification.

§4.2 Claims on the Three-Layer Structure

Claim 4.1 (Origin Sharing): For any curvature radius R, the subjective space $S^n(R)$ is defined by central projection centered at the **same origin** O of the background coordinates, by the definition in §3.1. Therefore, $S^n(R_1)$ and $S^n(R_0)$ share the common origin O.

Basis: Directly from the definition of central projection in [P1] §2. No new proof is needed.

Claim 4.2 (Projection-Axis Collinearity): Any half-line emanating from the origin O of the background coordinate space passes through both $S^n(R_1)$ and $S^n(R_0)$ in the **same direction**.

Basis: Directly from the background coordinate preservation property in [P1] §3. No new proof is needed.

Claim 4.3 (Concentric Nested Structure): From Claims 4.1 and 4.2, $S^n(R_1)$ and $S^n(R_0)$ are arranged as **concentric nested shells** sharing the same axial structure, centered at the origin O.

Basis: A trivial consequence of the combination of Claims 4.1 and 4.2.

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§5 Orthogonality of the R-Direction and Metric Non-Contribution

§5.1 Orthogonality of the R-Direction

Since $S^n(R)$ is a hypersphere of radius R centered at the origin O in the background coordinate space $\mathbb{R}^{n+1}$, at any point $p \in S^n(R)$ the radial direction (R-direction) is **orthogonal** to the tangent space $T_pS^n(R)$ of $S^n(R)$. Therefore, displacement in the R-direction **does not contribute** to the induced metric on $S^n(R)$.

§5.2 Metric Non-Contribution

Claim 5.1 (Metric Non-Contribution of R-Direction Displacement): On any half-line ℓ emanating from the origin O of the background coordinate space, a point p_1 on $S^n(R_1)$ and a point p_0 on $S^n(R_0)$ correspond to each other. The R-direction displacement connecting p_1 and p_0 is orthogonal to the tangent space of $S^n(R_0)$ and hence **does not contribute** to the geodesic metric on $S^n(R_0)$. Similarly, it does not contribute to the geodesic metric on $S^n(R_1)$. Therefore, the R-direction distance between p_1 and p_0 **cannot be measured by the intrinsic metric of any subjective space**.

Basis: An observation from the combination of the orthogonality in §5.1 and Claim 4.2. No new proof is needed.

Remark 1: This claim does not mean that p_1 and p_0 are the same point from the external viewpoint of the background coordinate space, or that the distance between them is zero. From the external viewpoint, they are at different radial positions. The claim is confined to the geometric fact that the R-direction displacement connecting p_1 and p_0 is orthogonal to the tangent spaces of the subjective spaces, and hence **the distance in that direction does not contribute to the metric**.

Remark 2: This claim does **not** mean that the values of the curvature radii R_0, R_1 themselves cannot be perceived from inside the subjective space. Since the curvature radius R determines the geodesic curvature of $S^n(R)$, a change in the value of R alters the internal geodesic structure (curvature distortion), which is perceptible to an internal observer. What this claim states is the more limited fact that the **R-direction distance** between corresponding points on different subjective spaces cannot be measured.

§5.3 Inter-Layer Map via Central Projection

For any point p_1 on $S^n(R_1)$, the map that assigns the intersection p_0 of the half-line from the origin O through p_1 with $S^n(R_0)$,

$$\Phi_{1\to 0} : S^n(R_1) \to S^n(R_0), \quad \Phi_{1\to 0}(p_1) = P_{R_0}(P_{R_1}^{-1}(p_1))$$

is well-defined and regular.

Basis: Directly from the central projection definition in §3.1 and the background coordinate preservation in §3.2.

Note: P_{R_1} is a surjection from $\mathbb{R}^{n+1} \setminus \{O\}$ to $S^n(R_1)$ and is not injective (all points on the same half-line project to a single point). Here $P_{R_1}^{-1}(p_1)$ refers to the entire half-line that projects to p_1. By the background coordinate preservation of §3.2, this half-line is uniquely recovered from p_1, and applying P_{R_0} yields the unique image p_0 on $S^n(R_0)$, so $\Phi_{1\to 0}$ is well-defined.

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§6 Summary of Claims

The geometric structure claimed by this paper is exhausted by the following.

1. Any subjective space defined by central projection shares the origin of the background coordinates (§4.2, Claim 4.1).
2. Subjective spaces of different curvature radii are arranged as a concentric nested structure (§4.2, Claim 4.3).
3. The R-direction displacement connecting corresponding points on different subjective spaces is orthogonal to the tangent spaces of all subjective spaces, and hence the distance in that direction cannot be measured by any intrinsic metric (§5.2, Claim 5.1).
4. The inter-layer map via central projection between subjective spaces is well-defined and regular (§5.3).

All of these are properties derived directly from the foundational papers [P1], [P2], [P8], and this paper bears no independent proof burden.

The originality of this paper lies in the **viewpoint of integrating these known properties as a middleware structure**. Namely, by hierarchizing R as a generic symbol and R_1 and R_0 as its instances, and positioning central projection as the interface connecting them, we obtain a **stable geometric framework independent of specific implementations of internal structure**.

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§7 Implementation Examples (Not Claims of This Paper)

The content of this section is not a claim of this paper but an example of concretizing the three-layer structure model. Even if the implementations mentioned in this section (6-dimensional hyperrectangle, soliton waves, micro black hole interpretation, etc.) are incorrect, the claims of §3–§6 are unaffected.

§7.1 Lower-Layer Implementation Example: Particle Internal Structure

An implementation can be considered in which $R = R_1$ is identified and each particle is treated as a closed geometric object possessing its own subjective space $S^n(R_1)$. In this implementation, the internal structure of a particle is described as a combinatorial structure inside $S^n(R_1)$. A concrete example is the 6-dimensional hyperrectangle model of [W8] (classification of 62 particles by axis orientation of xyztRQ).

This paper does not claim that this implementation is correct. The geometric structure of central projection holds equally whether the interior of a particle is a 6-dimensional hyperrectangle or any other combinatorial structure.

§7.2 Upper-Layer Implementation Example: Location of Phase Space

An implementation can be considered in which the phase space of a particle as a soliton wave is located inside $S^n(R_1)$. In this implementation, the position and phase information of each particle instance in a many-body problem is held inside its respective $S^n(R_1)$. On the $S^n(R_0)$ side, only observational information appears as an image via central projection.

This paper does not claim that this implementation is correct. As an alternative regarding the location of phase space, structurally one could also consider an arrangement in which the $S^n(R_0)$ side holds the phase space. The author considers the former arrangement, combined with the implementation of §7.1, to be more natural, but this preference is not a claim of this paper.

§7.3 Micro Black Hole Interpretation

An interpretation can be considered in which $S^n(R_1)$ is interpreted as a particle-scale micro black hole subjective space, and particle properties are described as an extension of the general-relativistic no-hair theorem. In this interpretation, to an internal observer of $S^n(R_0)$, the micro black hole appears as a central projection image (a point-like projection), and the internal structure is confined behind the event horizon.

This paper does not claim that this interpretation is correct. The interpretation is merely one reading of the geometric structure of central projection, and other interpretations are not excluded.

§7.4 Implementation Example of Interaction Mechanism

An implementation can be considered that interprets the metric non-contribution of the R-direction displacement in §5.2 as a geometric mechanism for interaction between different particles (different $S^n(R_1)$ instances). When two particle subjective spaces $S^n(R_1^{(A)})$ and $S^n(R_1^{(B)})$ have an R-direction connection that is unmeasurable by intrinsic metric on the same background coordinate half-line, an interaction channel opens between them.

This paper does not address the dynamical details of this implementation. The specific equations of interaction, coupling constants, and the role of mediating particles are deferred to future papers.

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§8 Conclusion

This paper is a middleware paper that organizes and integrates the geometric properties of subjective spaces established in the central projection series foundational papers [P1], [P2], [P8] as a three-layer structure R–R_1–R_0. Without proving new theorems, the following structure was observed as a combination of known properties:

- All subjective spaces share the common origin of the background coordinates
- Subjective spaces of different curvature radii are arranged as concentric nested shells
- The R-direction displacement connecting corresponding points on different subjective spaces is orthogonal to the tangent spaces of all subjective spaces, and hence the distance in that direction cannot be measured by intrinsic metric
- The inter-layer map via central projection between subjective spaces is well-defined and regular

This structure is independent of specific implementations such as particle internal structure and phase-space dynamics, and is positioned as a **middleware geometry** that remains stable even when the implementations of the upper and lower layers are changed.

In future papers, specific implementations of particle internals (6-dimensional hyperrectangle model, etc.) and phase-space dynamics (soliton wave model, etc.) will be built step by step on top of this middleware structure. Each implementation paper will be developed as an independent object of verification while presupposing the geometric foundation of this paper.

Position within the series: This paper is positioned between the central projection series foundational paper group (Papers 1–10) and the phase equation series (W1–W9). It integrates the individual geometric properties established by the foundational paper group into a three-layer structure and provides a stable interface on which the implementations of the phase equation series can rest.

§8.1 Explicit Statement of What This Paper Does Not Claim

To prevent misreading, the following are explicitly stated.

- **Does not claim the existence of micro black holes.** Although §7.3 mentions the micro black hole interpretation, this is one reading (implementation example) of the three-layer structure, and this paper does not contain the claim that micro black holes physically exist.
- **Does not claim many-worlds interpretation or many-worlds hypothesis.** The "nested structure" of this paper describes the geometric property of central projection that concentric spheres are arranged, and is unrelated to the many-worlds interpretation (Everett interpretation) in quantum mechanics or similar claims.
- **Does not claim that particles are condensed in the limit $R \to 0$.** This paper states that R_1 is smaller than R_0, but does not perform the limiting operation $R \to 0$. The picture of particles being condensed to a point is not included in the claims of this paper.

References

Cited in the main body (§3–§6)

- [P1]: Noriaki Kihara, Geometric Formulation of 4-Dimensional Space via Central Projection, Zenodo, 2026. DOI: 10.5281/zenodo.19427780
- [P2]: Noriaki Kihara, Geometric Symmetry of Central Projection: Mathematical Foundation of Multi-Axis Models, Zenodo, 2026. DOI: 10.5281/zenodo.19434932
- [P8]: Noriaki Kihara, Diameter of the Circumscribed Hypersphere of a Unit-Volume 4-Dimensional Hyperrectangle, Zenodo, 2026. DOI: 10.5281/zenodo.19533313

Cited in implementation examples (§7)

- [W8]: Noriaki Kihara, Set-Theoretic Structure and Combinatorial Properties of 6-Dimensional Hyperrectangles (v5), Zenodo, 2026. DOI: 10.5281/zenodo.19688521

Other

- Source code: https://github.com/WurabeSeiji/ai-chat-logs-open/

Appendix A: Self-Check on the Scope of This Paper

As a self-check of the writing policy, it is explicitly stated that the claims of this paper are unaffected in any of the following cases.

Hypothetical scenario that does not affect claims	Reason
The 6-dimensional hyperrectangle model [W8] is found to be entirely incorrect	§7.1 is an implementation example, independent of the claims in §3–§6

Hypothetical scenario that does not affect claims	Reason
The soliton wave model is found to be entirely incorrect	§7.2 is an implementation example, independent of the claims in §3–§6
The micro black hole interpretation is found to be physically inappropriate	§7.3 is an interpretation example, independent of the claims in §3–§6
Correspondence with individual particles in the Standard Model is found to be invalid	This paper does not claim individual particle correspondence

The claims of this paper hold in any of the above scenarios, as long as the foundational papers [P1], [P2], [P8] remain mathematically valid.

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v1.0